BANKHAUL · BUILD PLAN

BankHaul prototype build plan

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	BKH-BLD-001	Version	0.4
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/05-build-plan.md @ 659e1df		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: bank-edge board (O-2, line 18) made in section 3.8 and set in step 10, pull measured with and without it; far stake let down on a lowering rope in step 6 (O-7)
0.3	2026-10-25	Amish Chadha	Kit 1.9 D3 model; STEM fixes under Amish's standing instruction: pad eye 64 x 100 mm (D-S1), stays 1,110 mm between thimbles (D-S3), table legs, end rails and screws, cleat bolts M8 x 35 (D-S4); pictures redrawn from the D3 model
0.2	2026-10-03	Amish Chadha	Far stake 5.0 m long, driven 3.5 m (BKH-DDR-003, Amish's decision 14A); cost updated
0.1	2026-10-03	Amish Chadha	First build plan; design made constructable (BKH-DDR-002)

BankHaul prototype build plan

Plan, not yet built. How to build the first proof-of-concept station, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register (`docs/06-design-decisions.md`), not here.

> **Safety:** BankHaul is a rope system worked beside crocodile water. Placing the far stake is the most dangerous job in this plan: it is done from a boat or at low water, with a second person watching the water, never by wading. Ropes under load can snap back; nobody stands in the bight of the loop or in line with a loaded stay. Section 6 lists the points where work stops.

1. What you are building

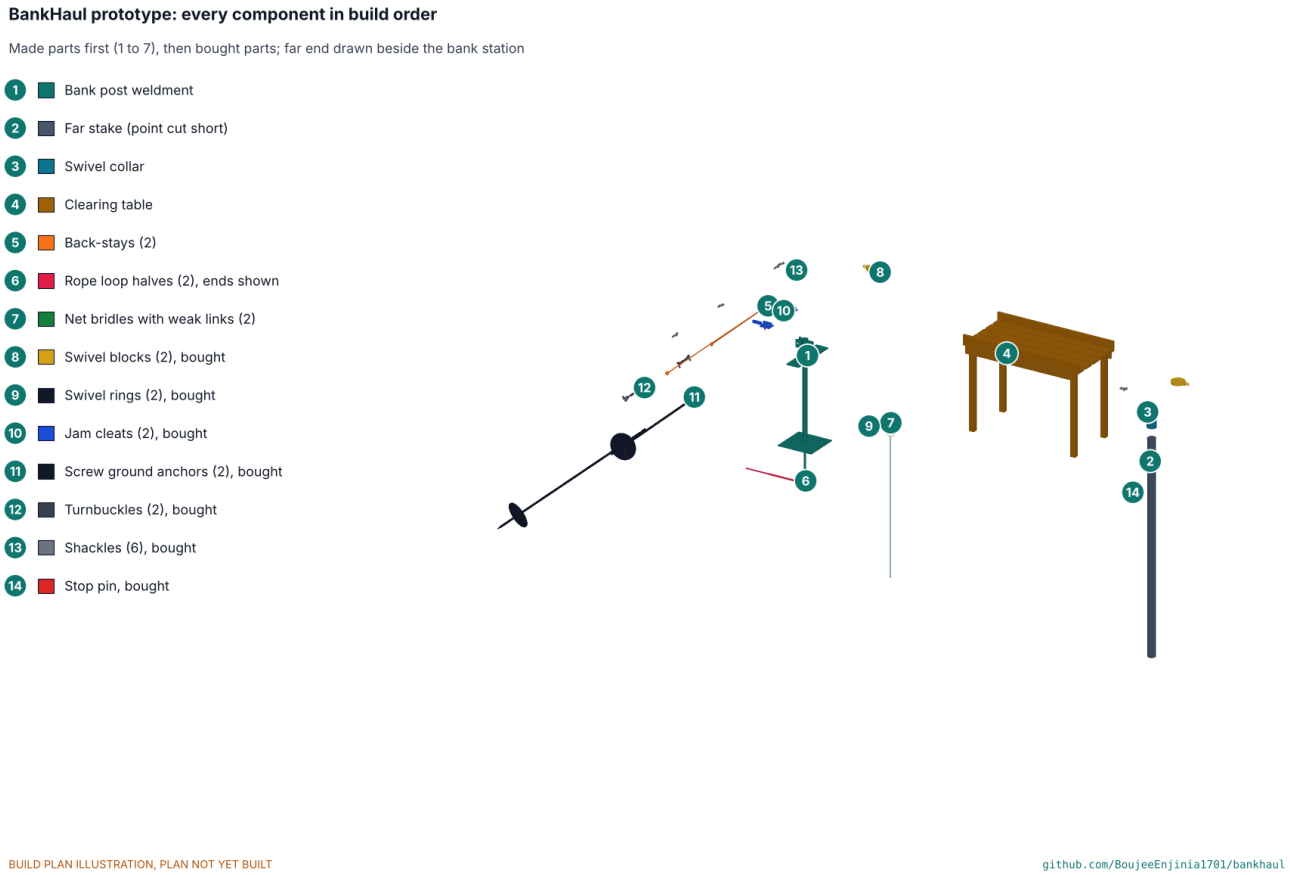


Figure 1. Every component of the prototype station, numbered in build order. The far end is drawn beside the bank station so all parts show.

The prototype is one net-hauling station for a 30 m span: a steel post on the bank held back by two rope stays to two screw ground anchors, a block at the post's head, a steel stake driven into the bed with a turning collar and a second block, a loop of floating rope between the blocks, two net bridles with weak links, and a timber clearing

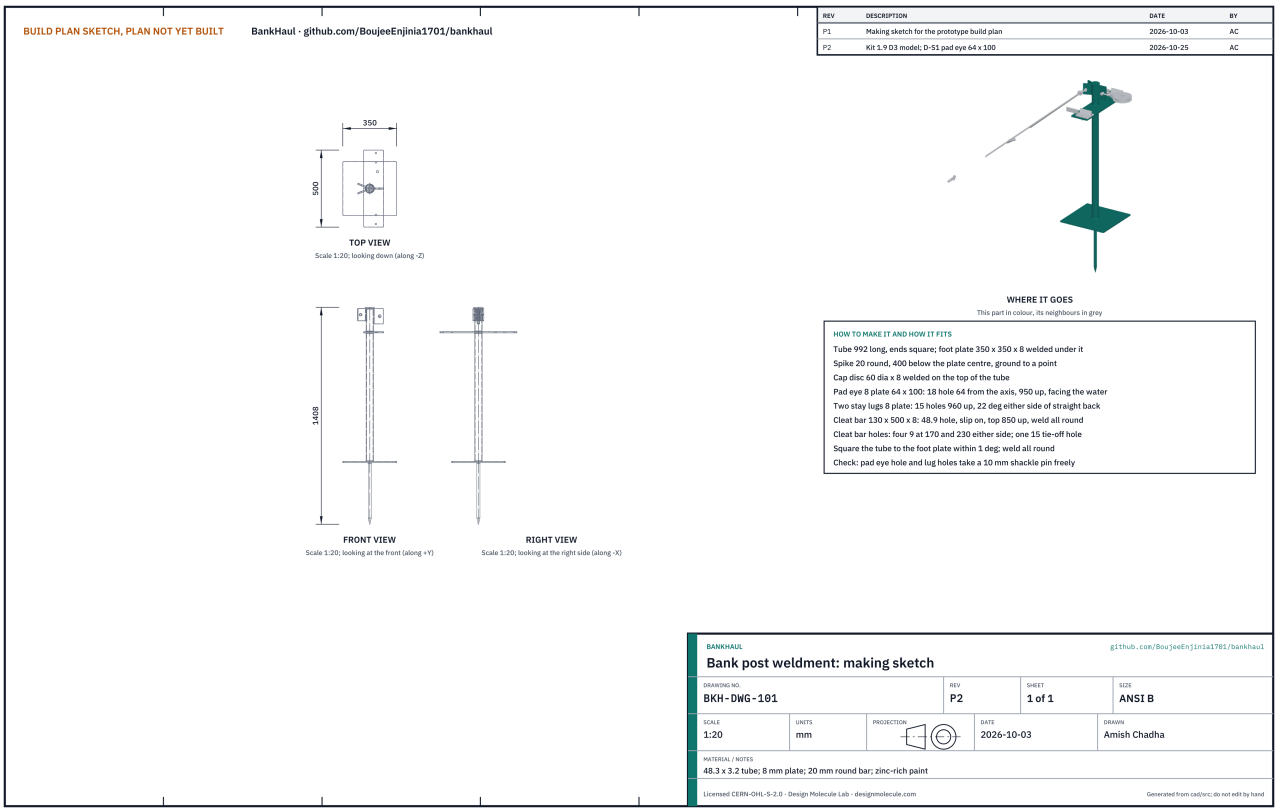
table. Seven components are made (the post, stake, collar and table by welding, drilling and carpentry; the stays, loop and bridles by rope work) and seven are bought (blocks, swivel rings, jam cleats, screw anchors, turnbuckles, shackles and a stop pin). The station parts cost about USD 399 from the bill of materials.

2. What changed to make it buildable

Component	The concept had	The buildable design has	Why
Bank post	A screw or deadman anchor carrying the pulley at waist height	A steel post on a foot plate and spike, held back by two splayed stays to two screw anchors	A post alone cannot hold 3 kN at waist height in soft soil; the stays take the pull as tension
Near pulley	A pulley on the anchor	A bought swivel block on a shackle through a pad eye at 950 mm	Commodity parts that line up with the loop by themselves
Cleat and brake	A cleat on the anchor	A cleat bar slipped over the post with a jam cleat for each leg and a tie-off hole	The round post has no flat face for a cleat
Screw anchors	Not defined	Two anchors with 250 mm helixes, screwed in line with the stays	The rods stay in tension; each holds about twice its load
Far stake	A steel or hardwood stake	A 76.1 mm steel tube with a welded point and cap, 5.0 m long, driven 3.5 m into the bed	A lighter stake bends at the design load; a shallower one can lean over in very soft mud
Far pulley and swivel	A pulley with a swivel	A collar that turns on the stake, resting on a stop pin that moves with the water level, with a bought block on a shackle	The collar is the swivel; the pin follows the season
Rope loop	An endless spliced loop	Two halves joined by two swivel rings, both ends of each half eye-spliced (the second at installation, D-S6)	The loop length can change with the shoreline; the rings are the clip points and end stops
Net clips	Quick clips	A bridle at each net end with a weak link and a snap hook	A seized net cannot pull more than about 700 N on the loop
Clearing table	A raised table or platform	A slatted timber table with lip boards, landward of the post	Keeps the net on the table, 5 m or more from the water

3. Making the components

3.1 Bank post weldment



What it is and what it is made from. The post that carries the near block, the jam cleats and the two stays. It is a 48.3 mm by 3.2 mm steel tube on a 350 mm square foot plate 8 mm thick, with a 20 mm round bar spike below, an 8 mm cap disc, an 8 mm pad eye, two 8 mm stay lugs and a 130 by 500 by 8 mm cleat bar. It weighs about 17 kg.

How to make it.

1. Cut the tube 992 mm long with square ends. Cut the foot plate 350 mm square and mark its centre.
2. Cut the spike 400 mm long and grind a point over the last 40 mm. Weld it square under the foot plate centre, all round.
3. Stand the tube on the plate centre, square within 1 degree in both directions, and weld all round. Weld the 60 mm cap disc on the top.
4. Cut the pad eye from 8 mm plate, 64 by 100 mm, so its end is 88 mm from the post's centre line, and drill an 18 mm hole 64 mm from the post's centre line and 950 mm above the ground. Weld it to the tube on the side that will face the water, top edge flush with the top of the tube.
5. Cut two stay lugs, 60 by 80 mm (ends 84 mm from the post's centre line), with 15 mm holes 960 mm above the ground. Weld them to the back of the tube, each turned 22 degrees off straight back, one to each side.
6. Cut the cleat bar 130 by 500 mm and cut a 48.9 mm hole 40 mm from its back edge. Drill four 9 mm holes for the cleats (170 mm and 230 mm either side of the post's centre line, 40 mm in front of it) and one 15 mm tie-off hole. Slide it down the tube until its top is 850 mm above the ground, square to the pad eye, and weld all round.
7. Clean and paint with zinc-rich paint.

How it fits the parts next to it.

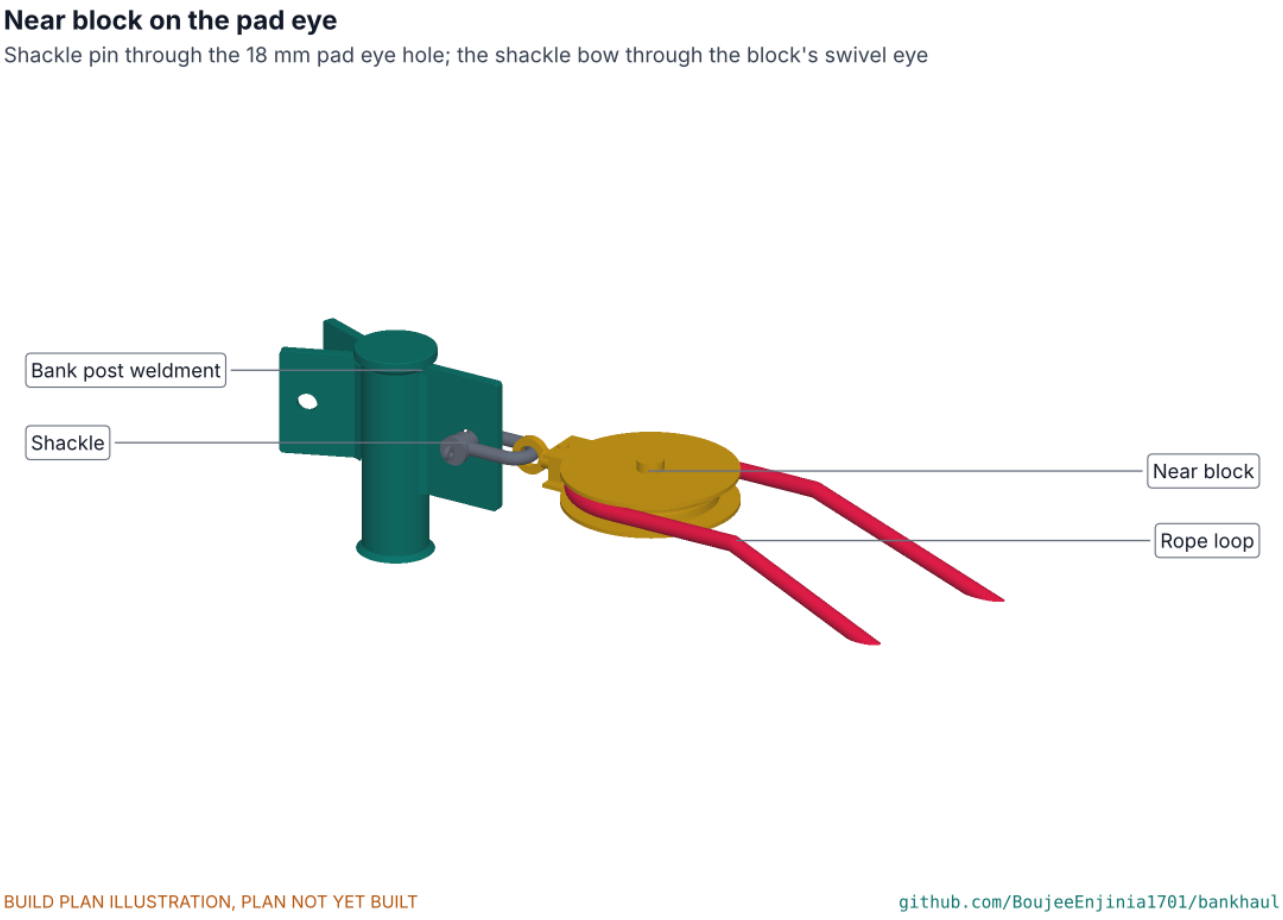


Figure 2. The near block hangs from the pad eye on a 10 mm shackle. The shackle pin passes through the 18 mm hole; the bow passes through the block's swivel eye. The block lies flat when the loop pulls on it.

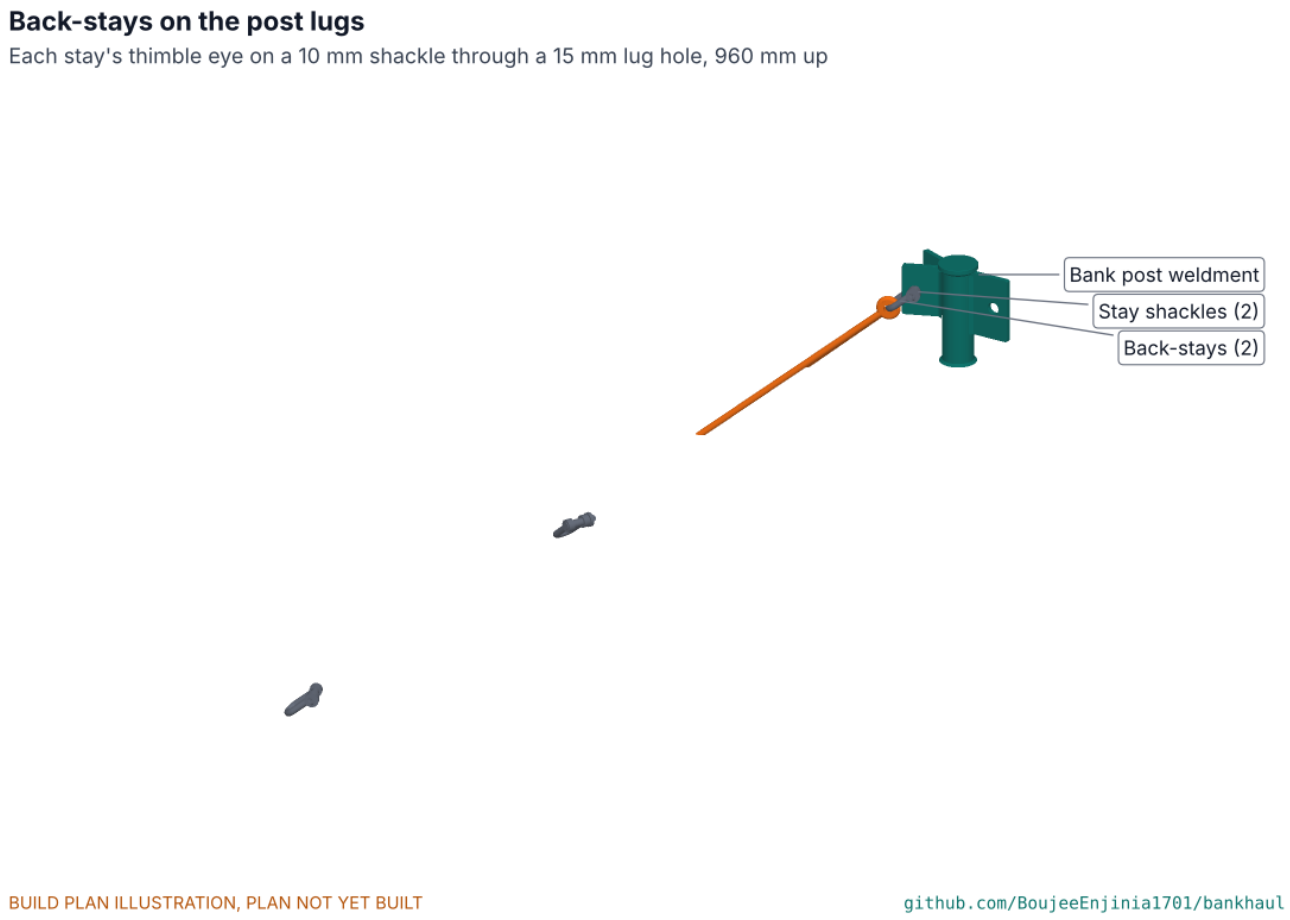


Figure 3. Each stay's thimble eye is held on a 10 mm shackle through a 15 mm lug hole. The lugs point along the stays, so the shackles pull straight.

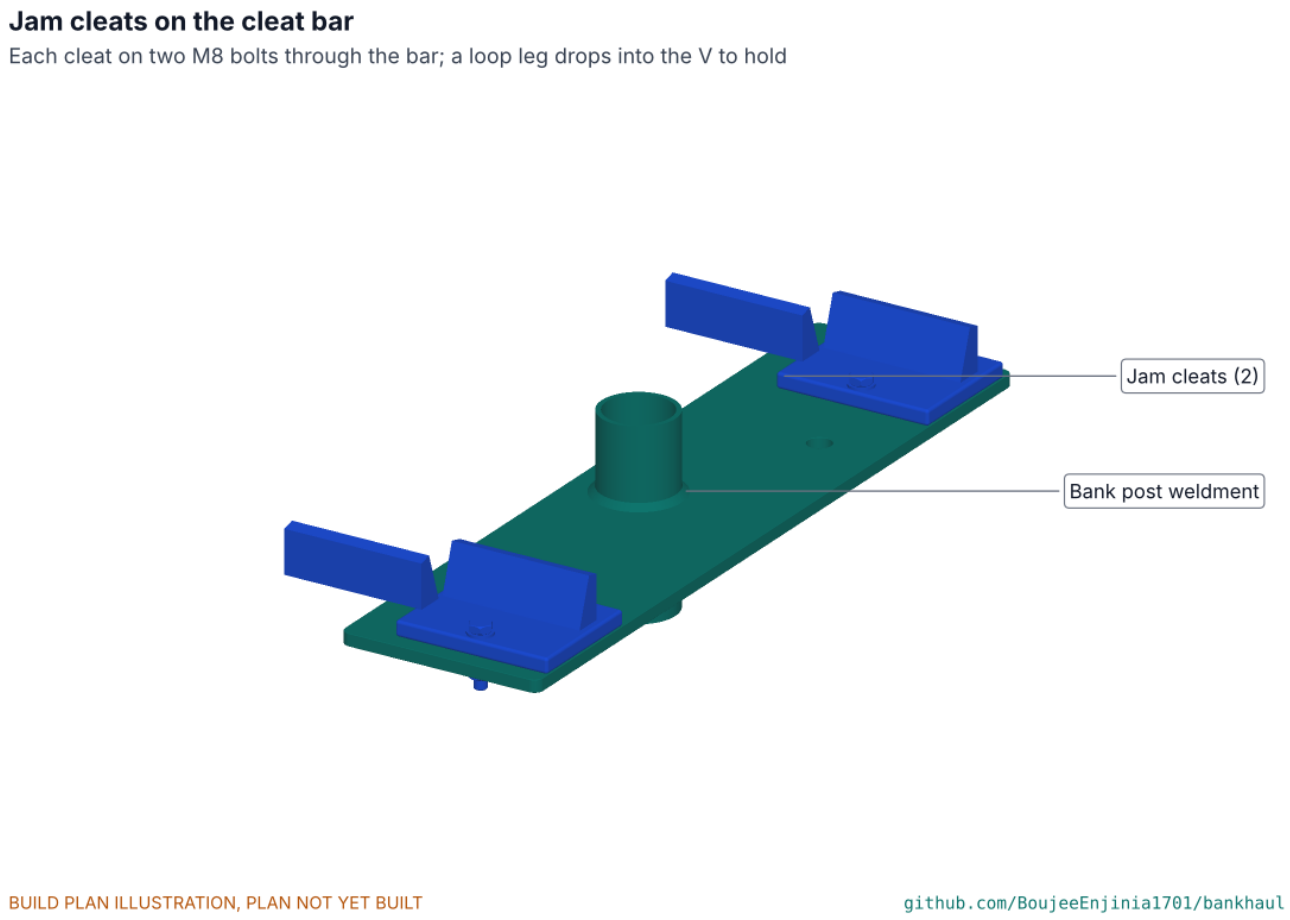


Figure 4. Each jam cleat sits on the cleat bar on two M8 bolts, its V pointing toward the water, beside one leg of the loop.

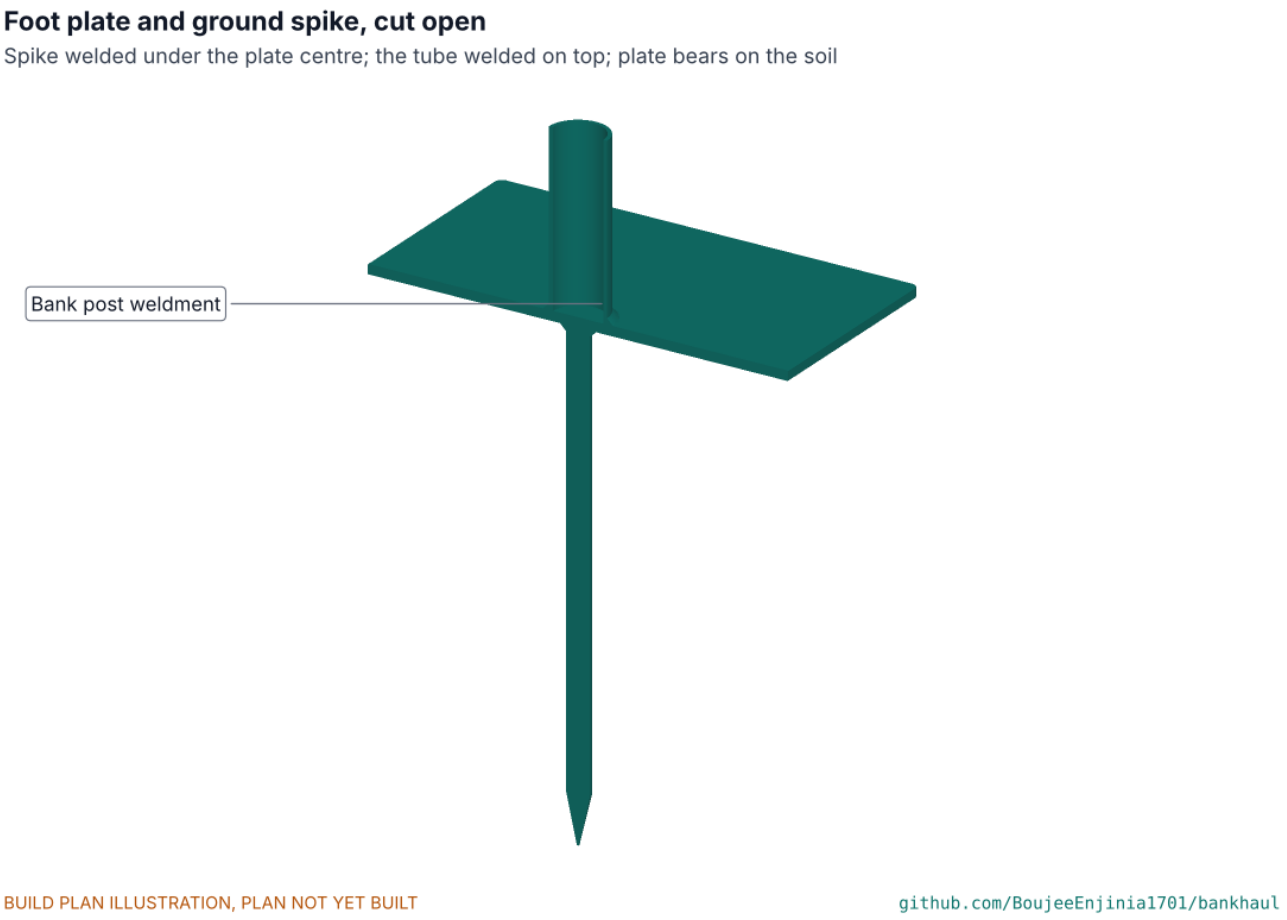
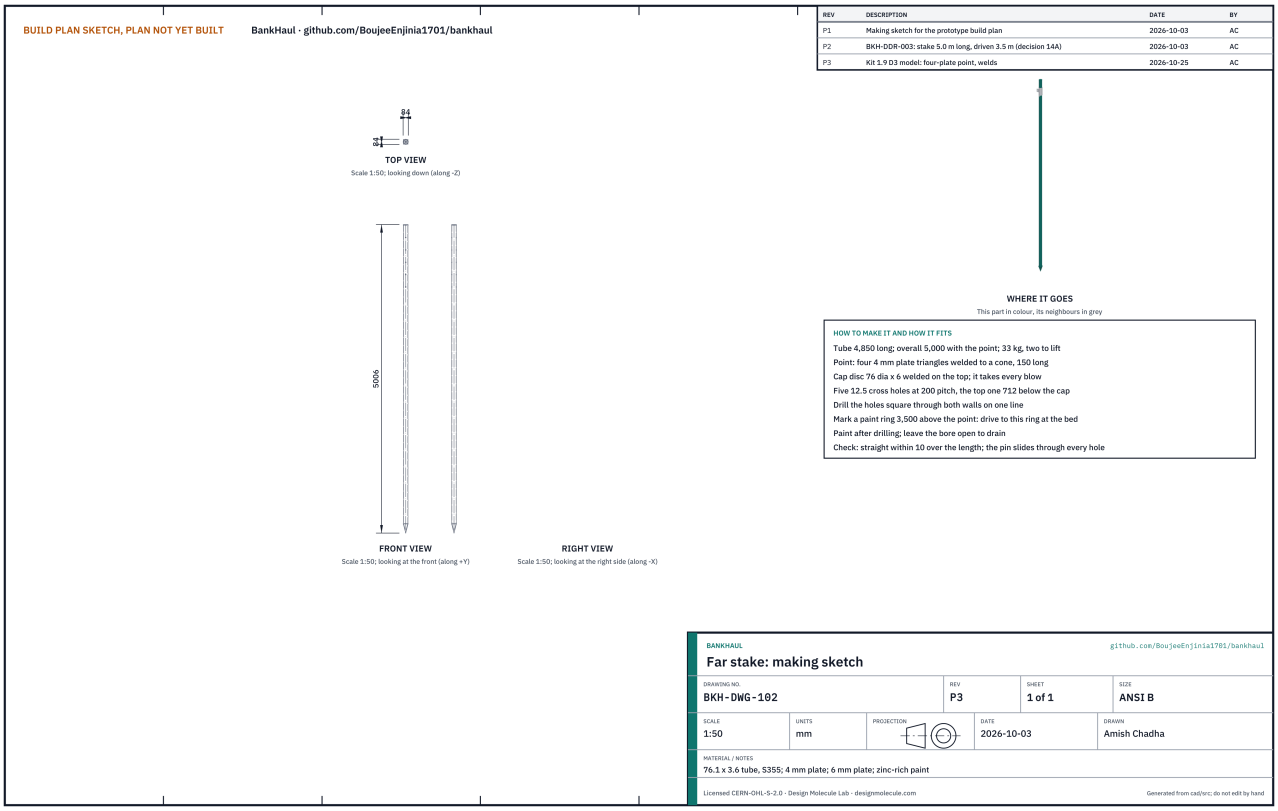


Figure 5. The spike is welded under the plate centre and the tube on top. The plate bears on the soil; the spike stops it sliding.

Check before moving on. A 10 mm shackle pin passes freely through the pad eye and both lug holes; the tube stands square on the plate; the cleat bar is level.

3.2 Far stake



What it is and what it is made from. The post in the bed that carries the far block. It is a 76.1 mm by 3.6 mm S355 steel tube, 5.0 m long overall with its point, with five cross holes for the stop pin. It weighs about 33 kg, so two people carry and lift it.

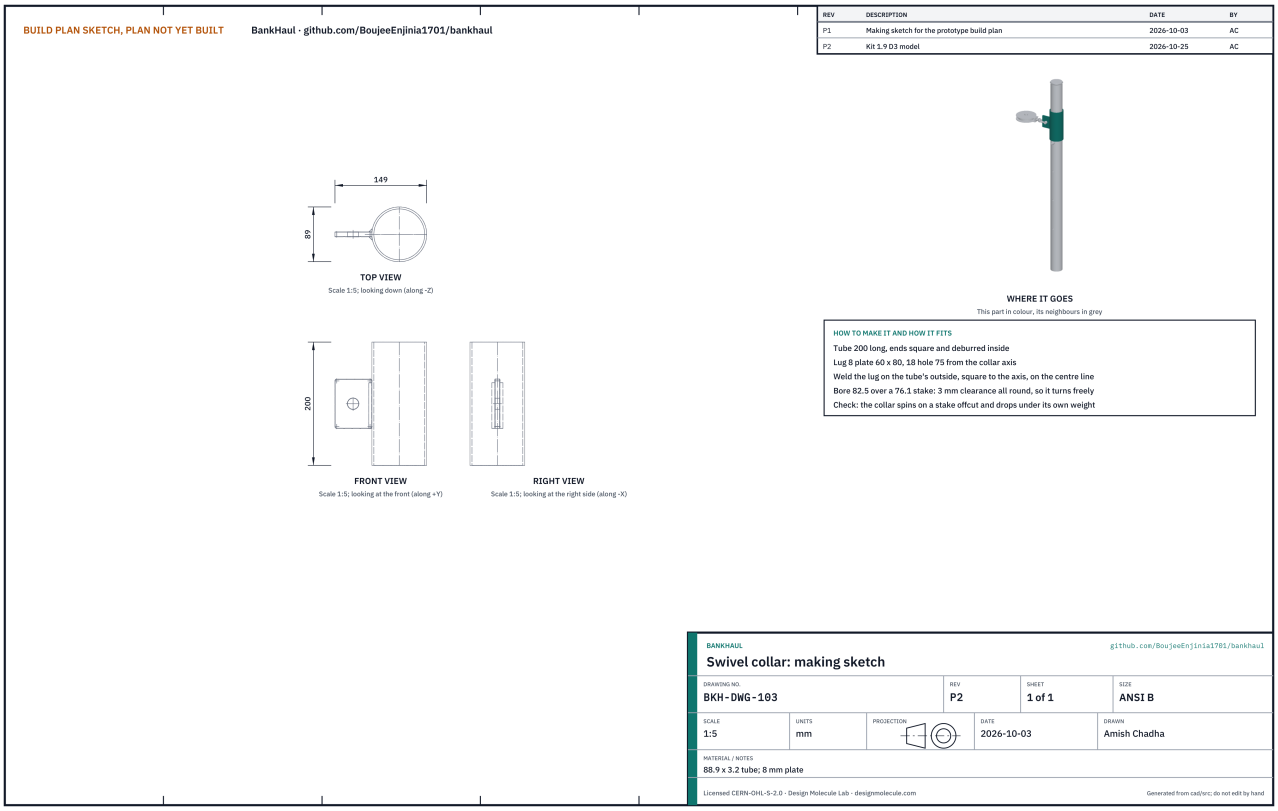
How to make it.

1. Cut the tube 4,850 mm long with square ends.
2. Cut four triangles from 4 mm plate and weld them into a cone point 150 mm long on one end. Weld a 6 mm cap disc on the other end; this disc takes every blow when driving.
3. Mark a paint ring 3,500 mm above the point: the stake is driven until this ring reaches the bed.
4. Drill five 12.5 mm holes straight through both walls on one line, 200 mm apart, the top one 712 mm below the cap.
5. Paint with zinc-rich paint after drilling.

How it fits the parts next to it. The collar slides over the top and rests on the stop pin (section 3.3, Figure 7).

Check before moving on. The stake is straight within 10 mm over its length, and a 12 mm bolt slides through every hole.

3.3 Swivel collar



What it is and what it is made from. A short tube that turns on the stake and carries the far block. It is an 88.9 mm by 3.2 mm steel tube, 200 mm long, with an 8 mm lug.

How to make it.

1. Cut the tube 200 mm long with square ends and remove the burr inside.
2. Cut the lug from 8 mm plate, 60 by 80 mm, with an 18 mm hole 75 mm from the collar's centre line.
3. Weld the lug to the outside of the tube, square to the tube's axis and centred on its length. Paint.

How it fits the parts next to it.

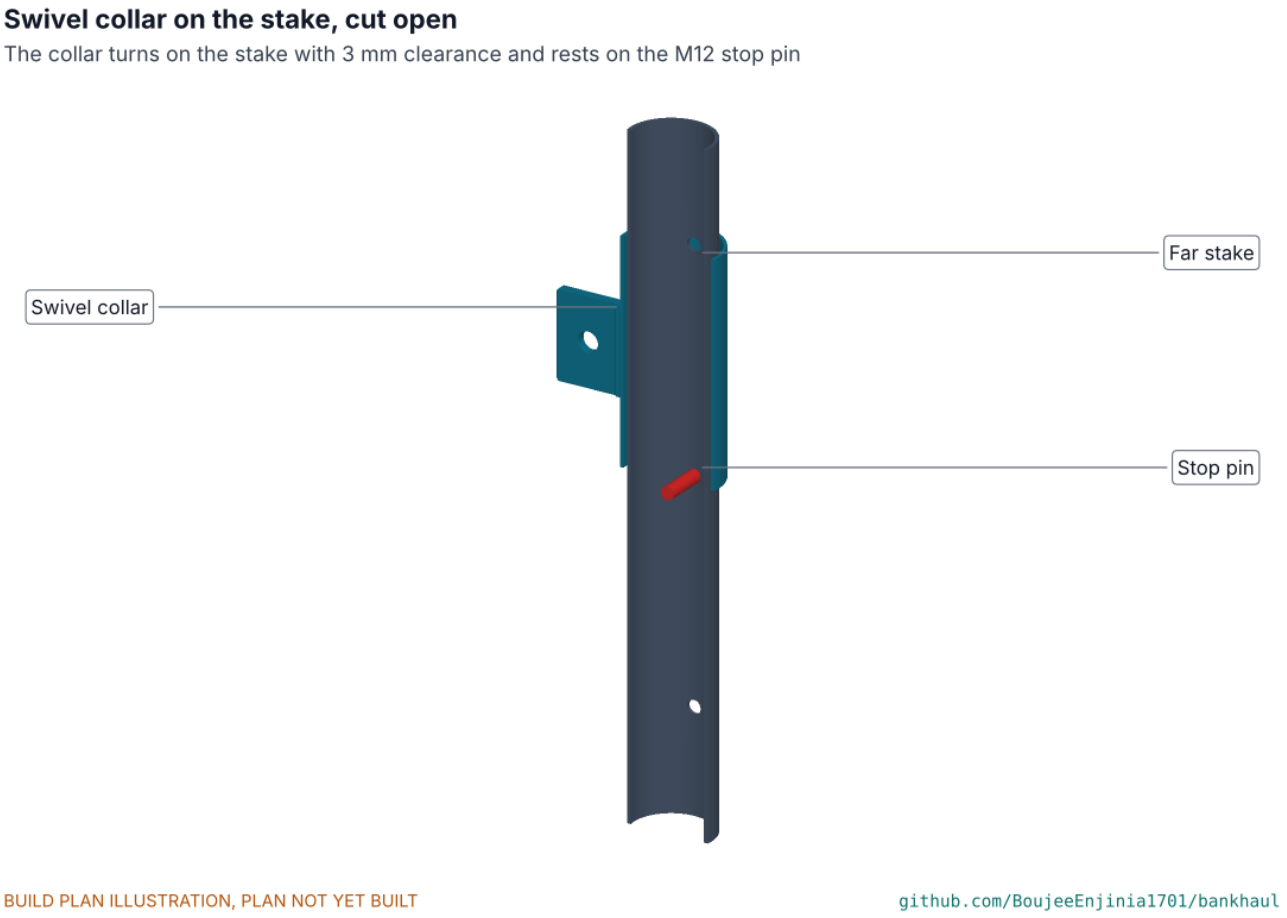
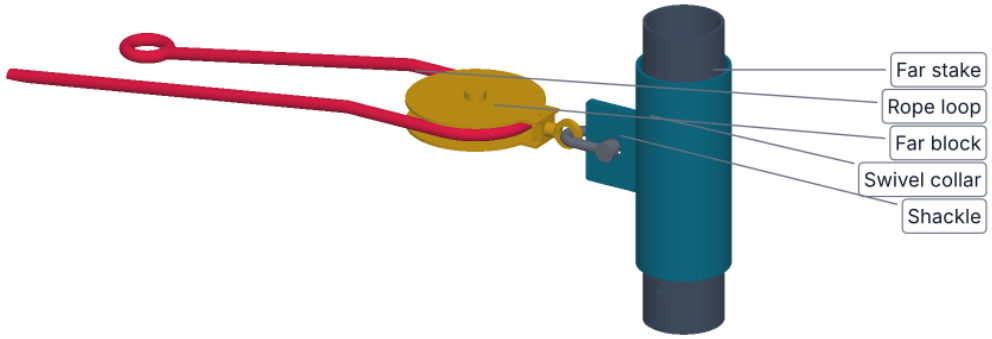


Figure 6. The collar turns on the stake with about 3 mm clearance all round and rests on the M12 stop pin through the stake.

Far block on the collar lug

Shackle pin through the 18 mm lug hole; the bow through the block's swivel eye; the loop round the sheave



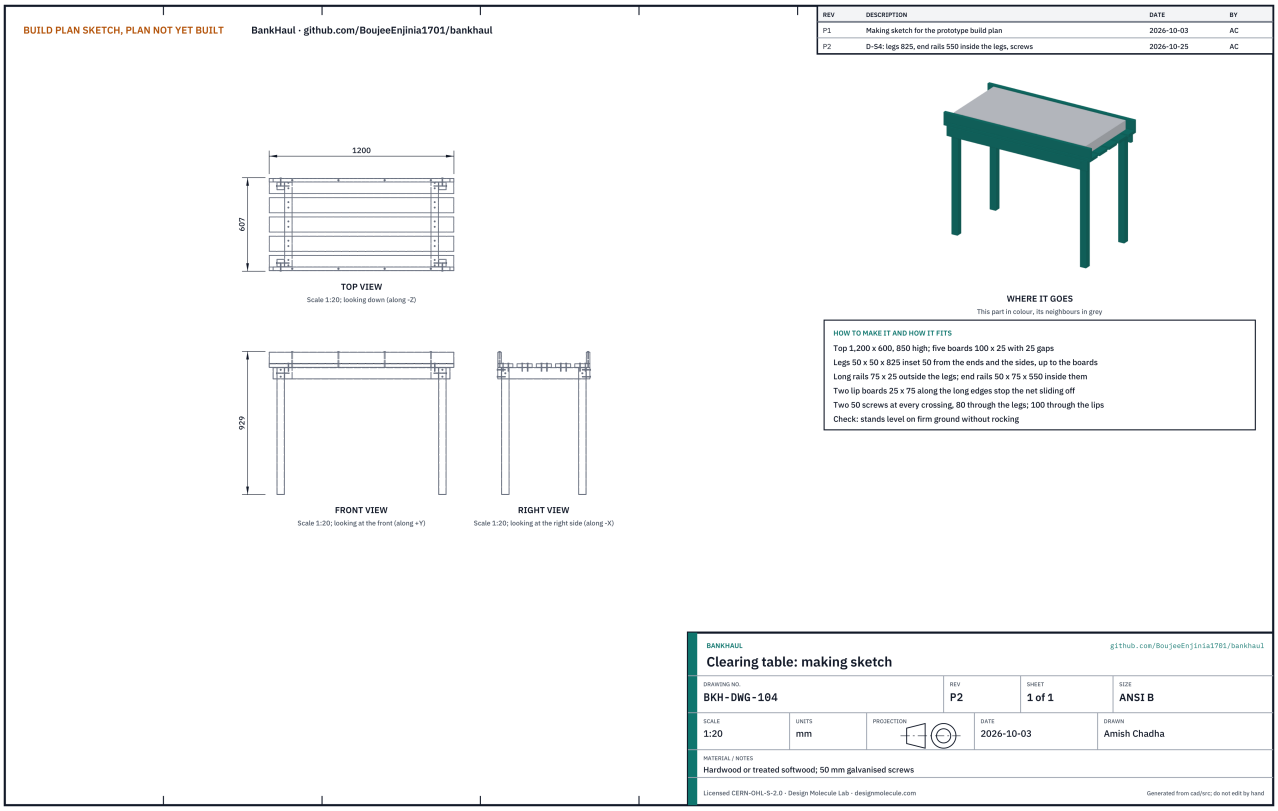
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/bankhaul

Figure 7. The far block hangs from the collar lug on a 10 mm shackle; the loop runs round its sheave and back toward the bank.

Check before moving on. The collar spins on an offcut of the stake tube and drops down it under its own weight.

3.4 Clearing table



What it is and what it is made from. A raised table on the bank where the net is stacked and cleared. Hardwood or treated softwood, held with 50 mm galvanised screws; the top is 1,200 by 600 mm and 850 mm high.

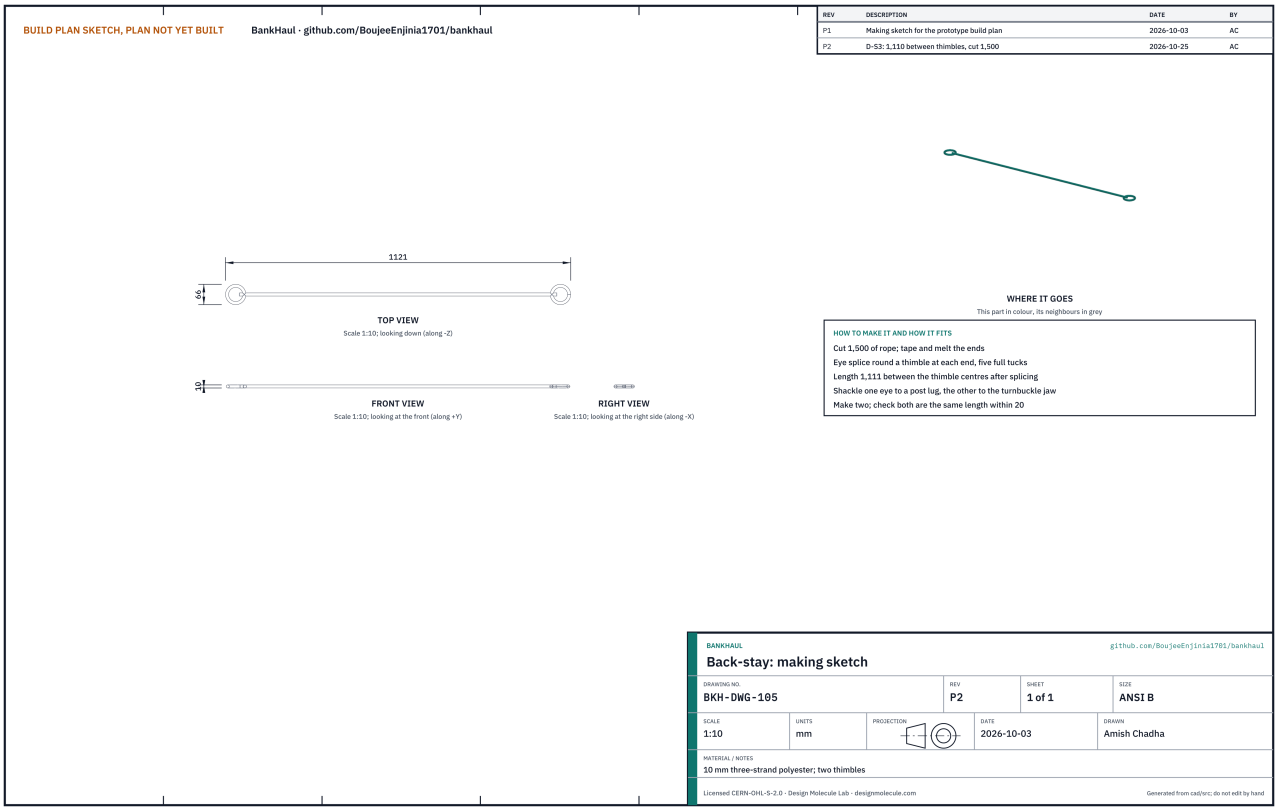
How to make it.

1. Cut four legs 50 by 50 by 825 mm, two long rails 75 by 25 by 1,150 mm, two end rails 50 by 75 by 550 mm, five top boards 100 by 25 by 1,200 mm and two lip boards 25 by 75 by 1,200 mm.
2. Screw the long rails to the outside of the legs, flush with the legs' tops, with two 50 mm screws at each leg; fit the end rails inside the legs, between the long rails, with two 80 mm screws through each leg.
3. Screw the top boards across the rails with 25 mm gaps so water drains.
4. Screw the lip boards on edge along both long sides with four 100 mm screws each, down through the outer top board.

How it fits the parts next to it. It stands free on the bank, landward of the post and on the side away from the stays.

Check before moving on. It stands level on firm ground without rocking.

3.5 Back-stays



What it is and what it is made from. Two short ropes that hold the post back. Each is 10 mm three-strand polyester with a thimbled eye splice at each end.

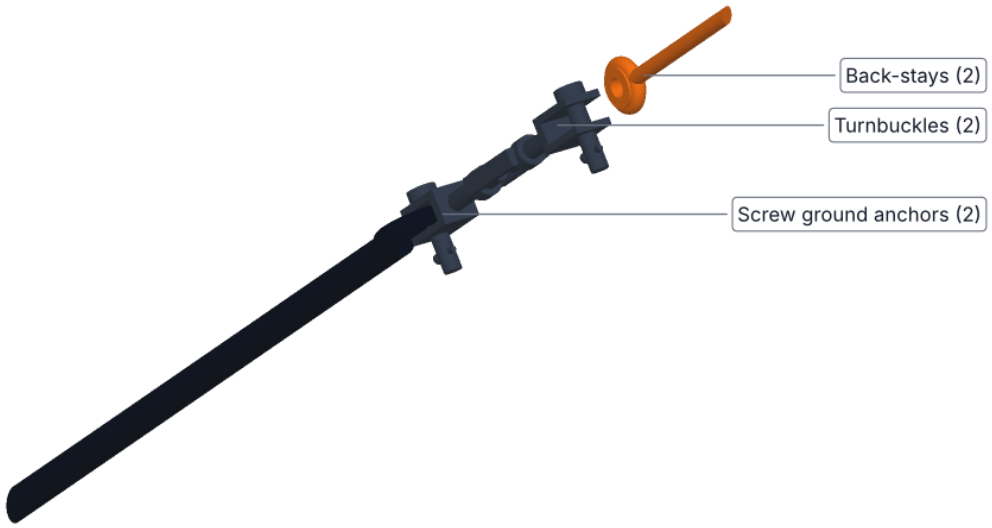
How to make it.

1. Cut two lengths of 1,500 mm; tape and melt the ends.
2. Splice an eye round a thimble at each end with five full tucks, so the thimble centres are 1,110 mm apart. With the two shackles and the turnbuckle at mid take-up, this fills the 1,440 mm from the lug hole to the anchor eye.

How it fits the parts next to it. One eye is shackled to a post lug (Figure 3); the other is shackled to the turnbuckle on the anchor eye.

Screw anchor, turnbuckle and stay

Turnbuckle jaw on the anchor eye; the stay's eye on the other jaw by a shackle; rod in line with the stay



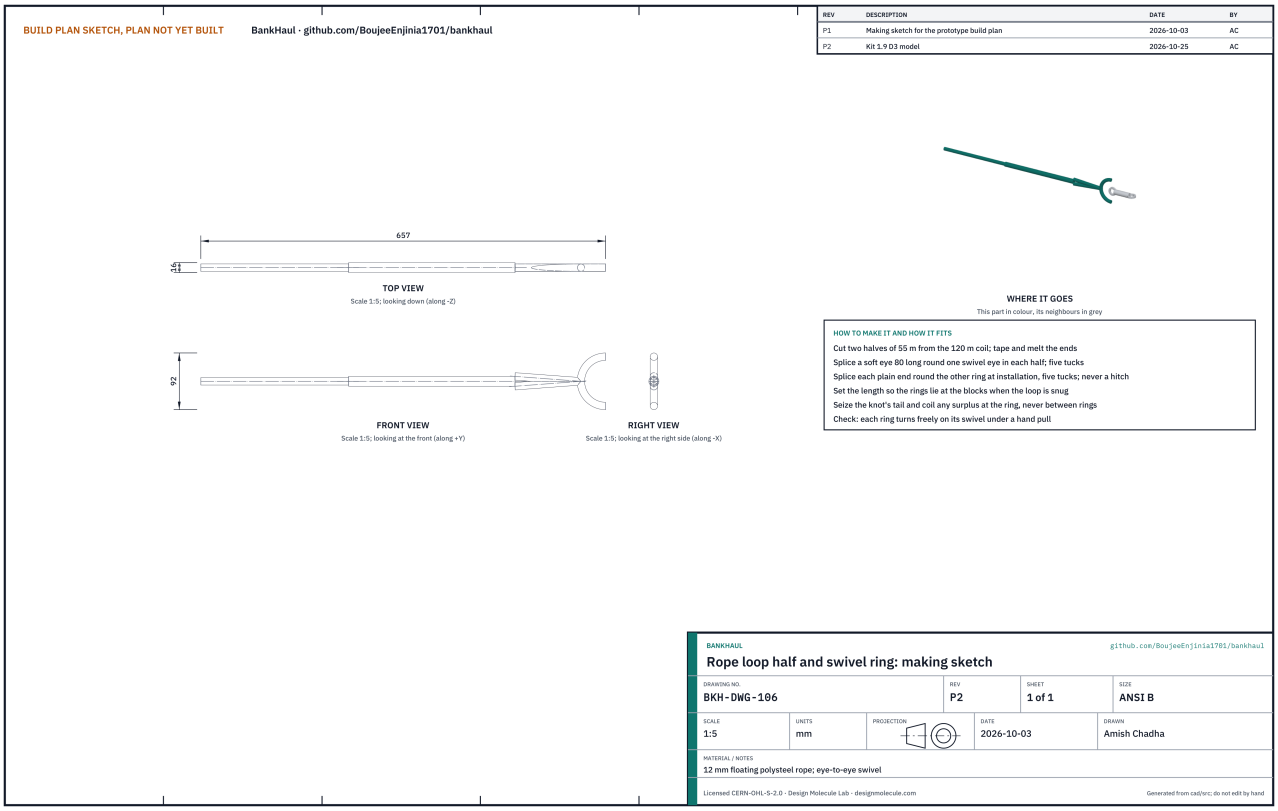
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/bankhaul

Figure 8. The turnbuckle's jaw takes the anchor's eye; the stay's eye is shackled to the other jaw. The anchor's rod runs in line with the stay.

Check before moving on. Both stays are the same length within 20 mm.

3.6 Rope loop



What it is and what it is made from. The loop that carries the net out and back: two halves of 12 mm floating polysteel rope joined by two eye-to-eye swivels, the swivel rings.

How to make it.

1. Cut two halves of 55 m from the 120 m coil; tape and melt all ends.
2. Splice a soft eye, about 80 mm long, round one eye of a swivel ring at one end of each half, with five tucks.
3. The plain end of each half is spliced round the other ring when the loop is reeved (step 9), at the length that suits the span. Never tie it with a hitch: a hitch keeps only about 60 % of the rope's strength (D-S6).

How it fits the parts next to it.

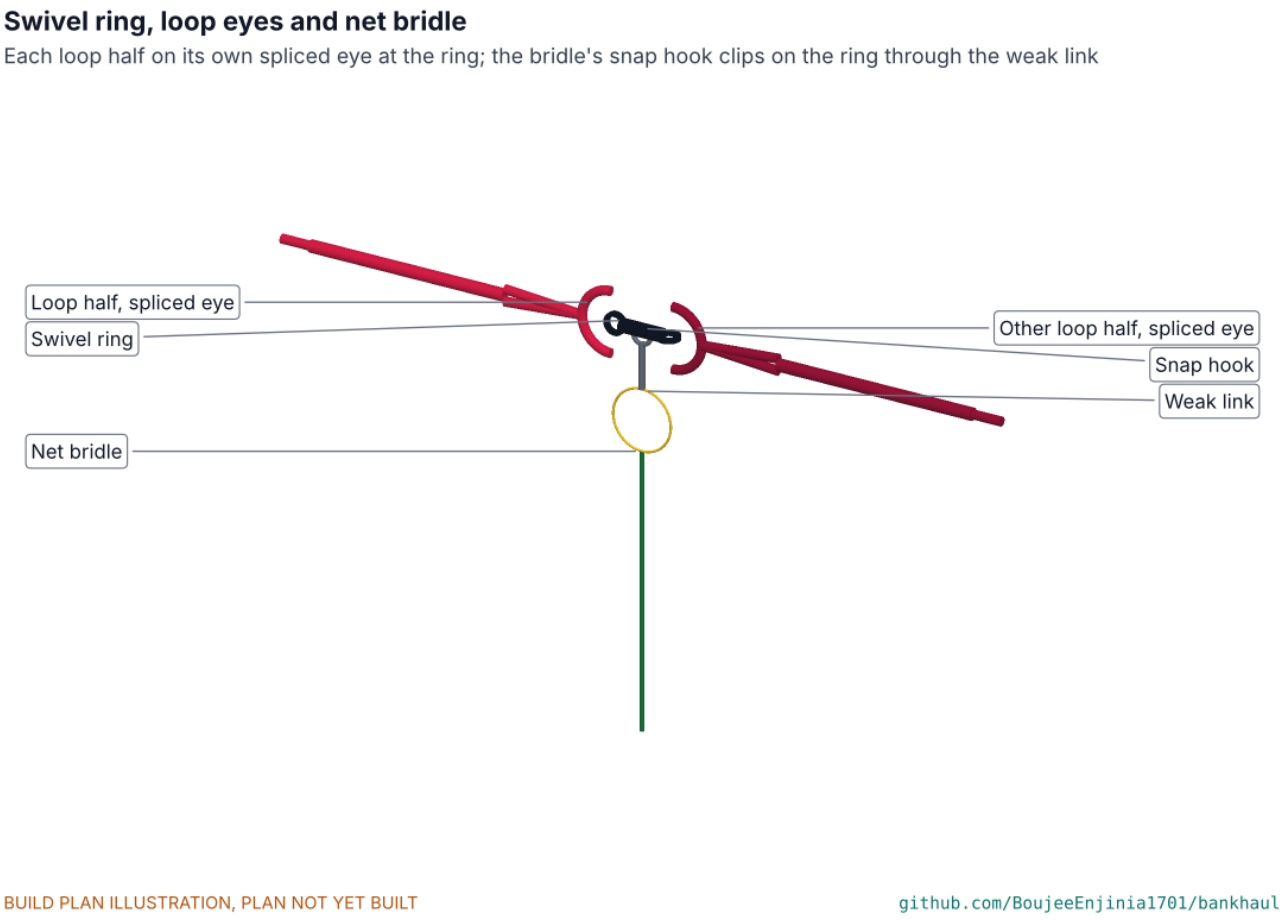
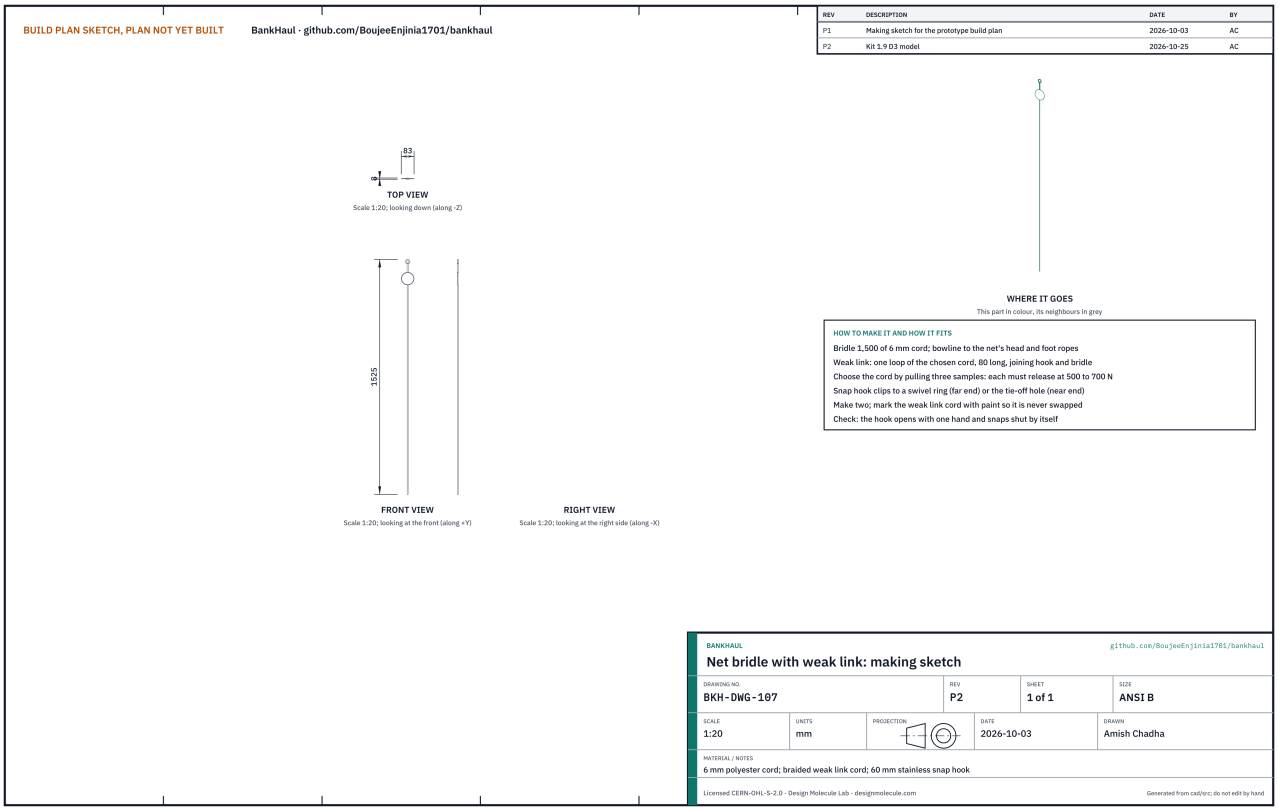


Figure 9. The two loop halves meet at each swivel ring, each on its own spliced eye. The net bridle's snap hook clips onto ring 1.

Check before moving on. Each ring turns freely under a firm hand pull; each splice has five full tucks.

3.7 Net bridles with weak links



What it is and what it is made from. One bridle for each end of the net: 1.5 m of 6 mm polyester cord, a weak link of braided cord and a 60 mm stainless snap hook.

How to make it.

1. Pull three samples of the weak link cord to breaking on a spring scale or the CalRig rig; use the cord only if every sample breaks between 500 and 700 N, as a single loop 80 mm long.
2. Tie the bridle cord to the head rope and foot rope at the net's end with bowlines.
3. Join the bridle to the snap hook with one loop of the weak link cord. Mark the link with paint so it is never replaced with stronger cord.

How it fits the parts next to it. The far-end bridle clips onto ring 1 (Figure 9); the near-end bridle clips into the tie-off hole of the cleat bar.

Check before moving on. The hook opens with one hand and snaps shut by itself.

3.8 Bought components

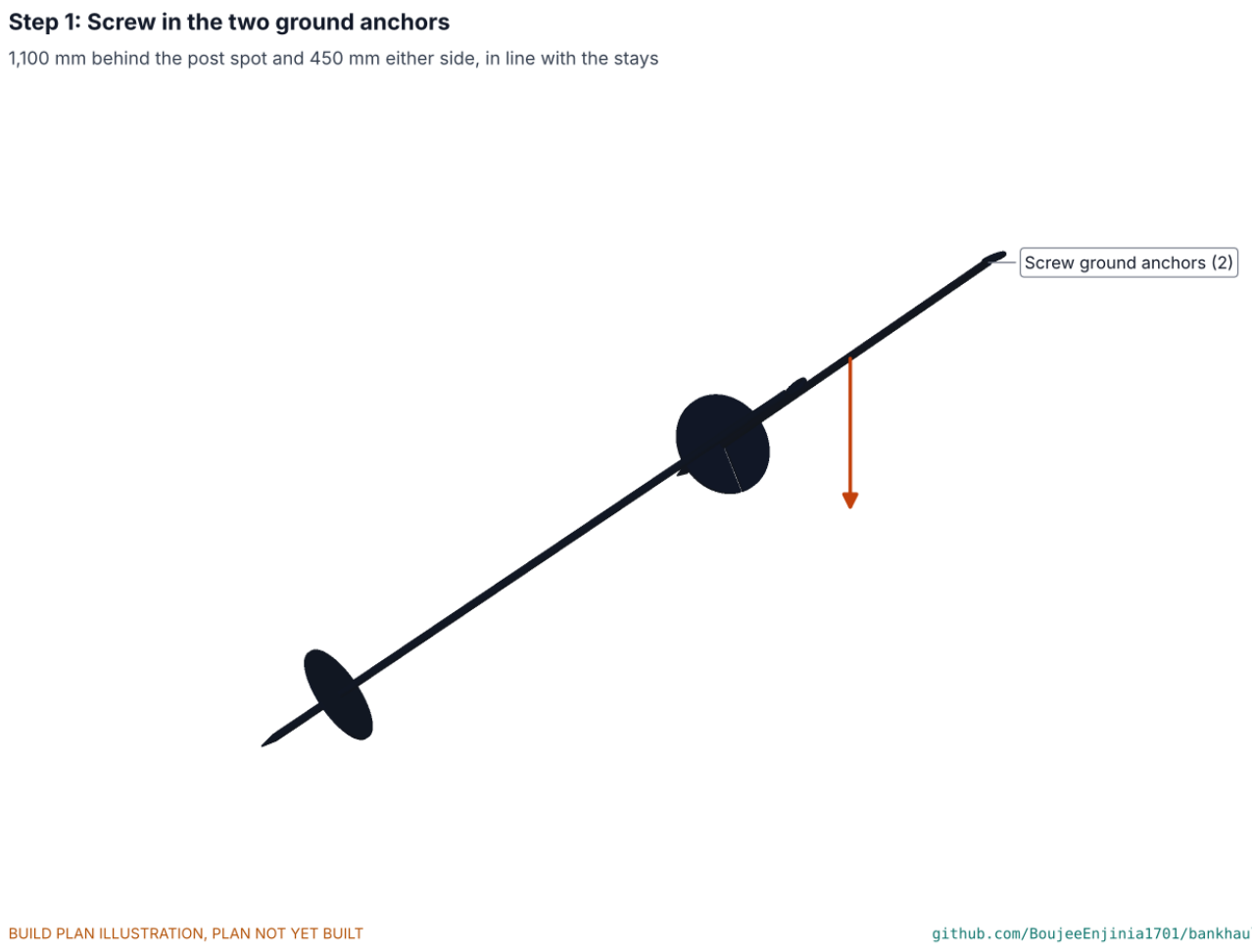
- **Swivel blocks (2).** Single sheave, 100 mm sheave for 12 mm rope, swivel eye, working load limit at least 500 kg. The eye and the spacer between the cheeks must sit outside the rope's path round the sheave. Nothing to do to them.
- **Swivel rings (2).** Eye-to-eye swivels, 10 mm, working load limit at least 500 kg, with eyes at least 25 mm inside.
- **Jam cleats (2).** V-jam cleats for 10 to 14 mm rope with two 8 mm fixing holes about 60 mm apart. If the hole spacing differs, drill the cleat bar to suit.

- **Screw ground anchors (2).** A single 250 mm helix on a 20 mm galvanised rod, 1.6 m long, with an eye.
- **Turnbuckles (2).** Rated M12 jaw-and-jaw, working load limit at least 500 kg.
- **Shackles (6).** Rated 10 mm bow shackles, working load limit at least 750 kg; wire every pin shut (mousing).
- **Stop pin.** An M12 by 100 mm stainless bolt with a nyloc nut and two washers, plus a spare.
- **Bank-edge board (made, line 18).** A smooth hardwood board 250 x 600 x 25 mm, every edge rounded to about 10 mm, planed smooth on top. It is set flush into the bank lip under both loop legs (step 10), so the catch and weed slide over the edge instead of dragging in the mud (BKH-DEC-001 O-2).
- **Stake driving cap (tool).** A 300 mm length of 88.9 mm tube with a 10 mm cap plate and two handles, made to fit over the stake top.

4. Putting it together

Steps 1 to 5 and 10 are done on the bank; steps 6 to 9 need a boat or a low-water day and two people.

Step 1: Screw in the two ground anchors

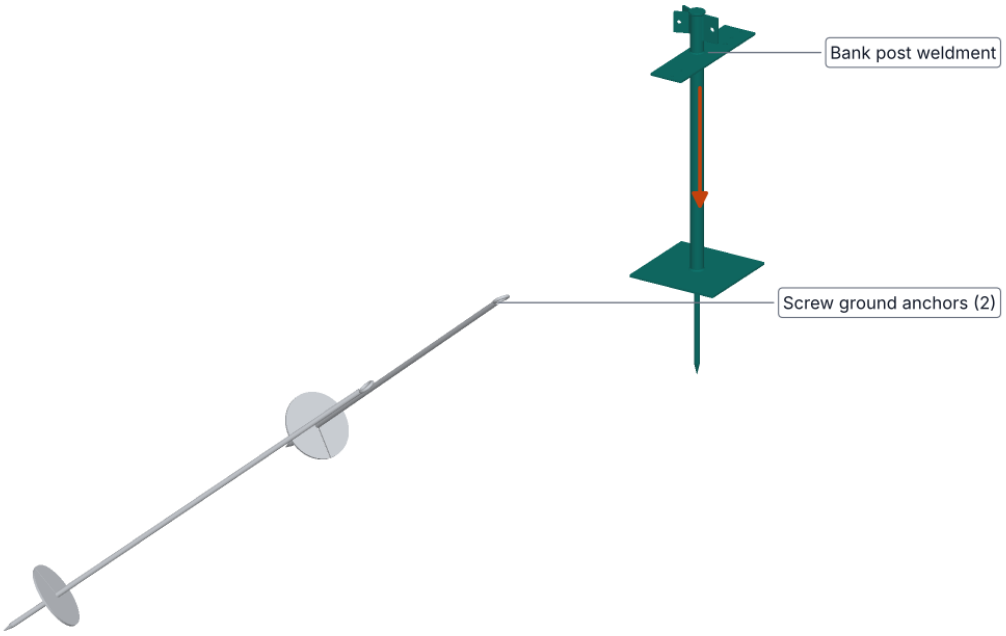


Mark the post spot at least 6 m back from the water's edge. Mark the two anchor spots 1,100 mm behind it and 450 mm either side of the line to the far stake. Screw each anchor in with a bar through its eye, leaning back toward the post at about 39 degrees from the ground, until the eye is just above the ground.

Step 2: Stand the post on its spike

Step 2: Stand the post on its spike

Push the spike into the ground until the foot plate sits flat; pad eye toward the water



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

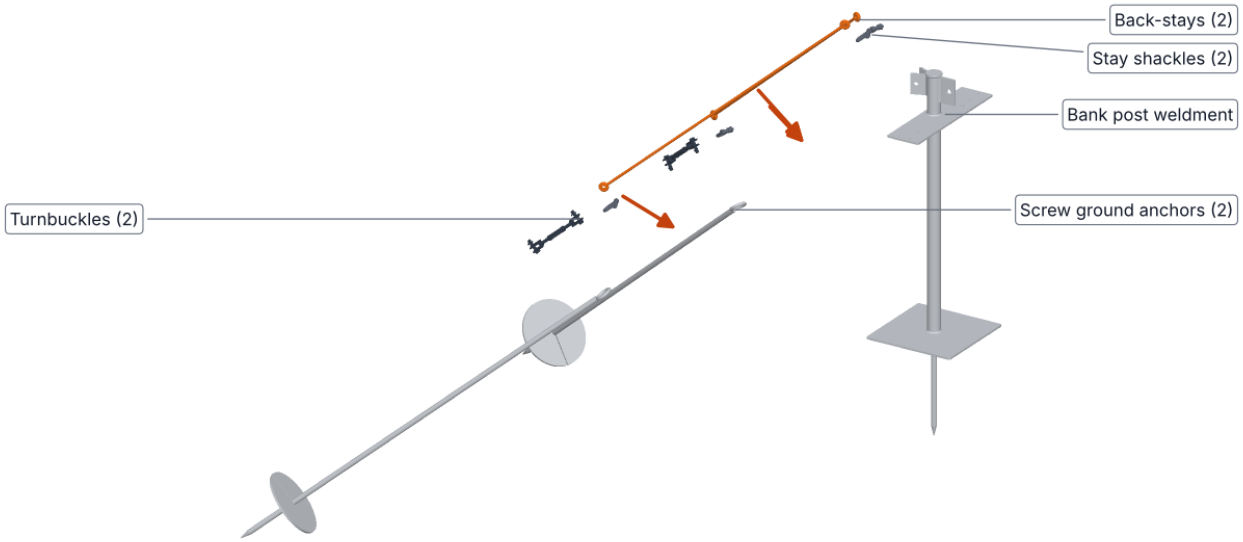
github.com/BoujeeEnjinia1701/bankhaul

Push the spike in until the foot plate sits flat, pad eye toward the water.

Step 3: Fit the back-stays

Step 3: Fit the back-stays

Shackle each stay to a lug; turnbuckle to the anchor eye; take up until the post leans 1 deg landward



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

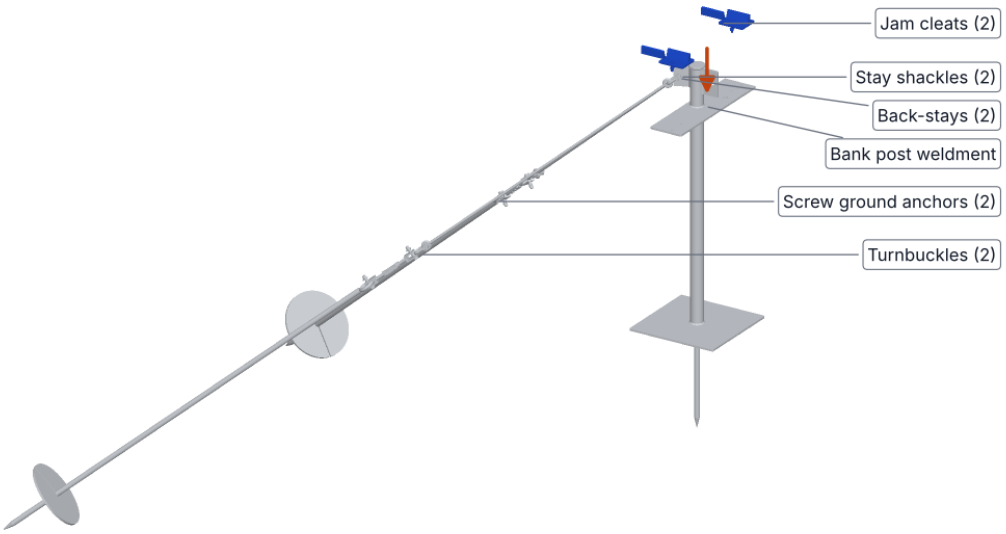
github.com/BoujeeEnjinia1701/bankhaul

Shackle each stay to a post lug and to a turnbuckle on an anchor eye. Take up both turnbuckles evenly until the post leans about 1 degree away from the water. Wire the shackle pins.

Step 4: Bolt on the jam cleats

Step 4: Bolt on the jam cleats

Two M8 bolts each, V toward the water



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

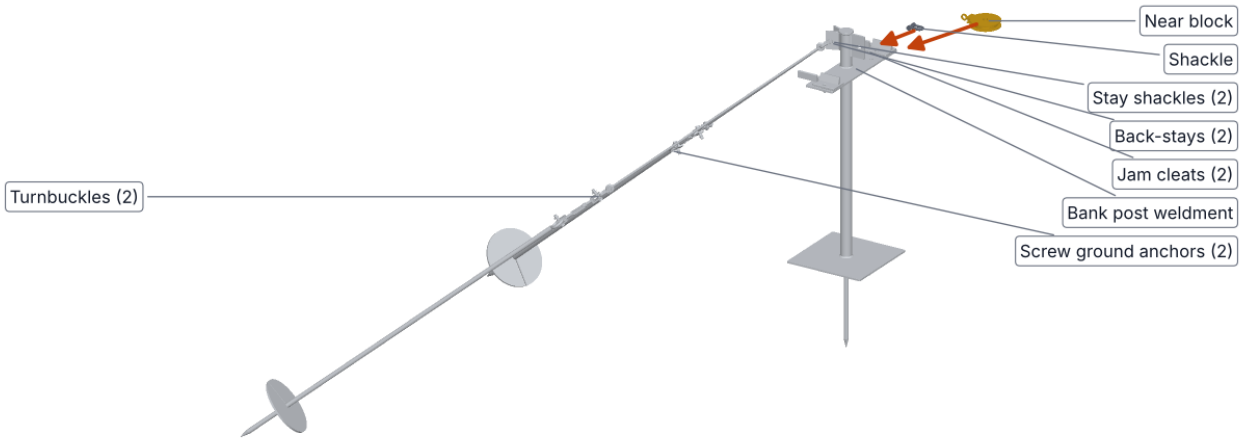
github.com/BoujeeEnjinia1701/bankhaul

Two M8 x 35 bolts each, with washers and nyloc nuts, the V pointing toward the water.

Step 5: Hang the near block

Step 5: Hang the near block

Shackle through the pad eye and the block's swivel eye; mouse the pin

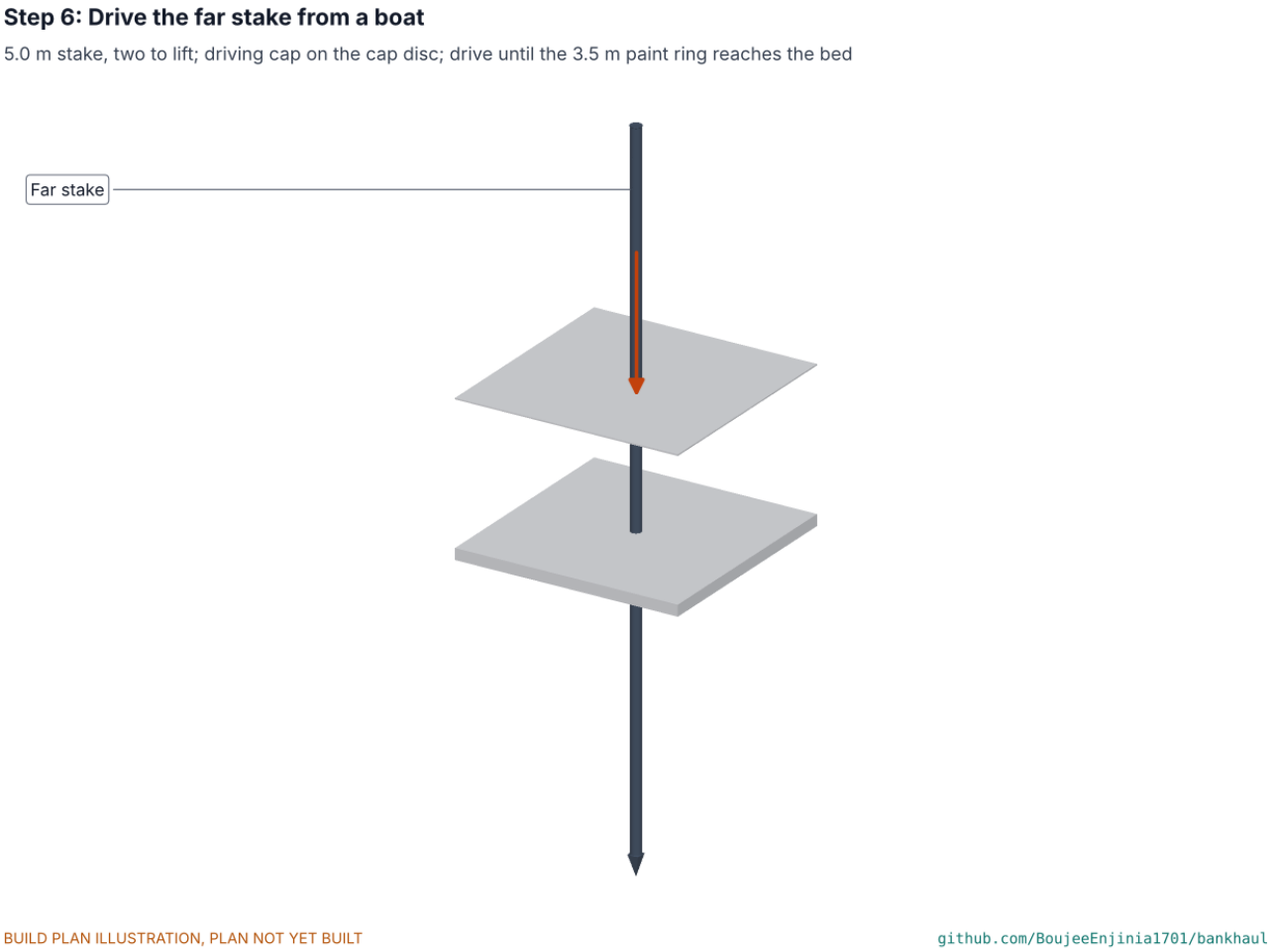


BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/bankhaul

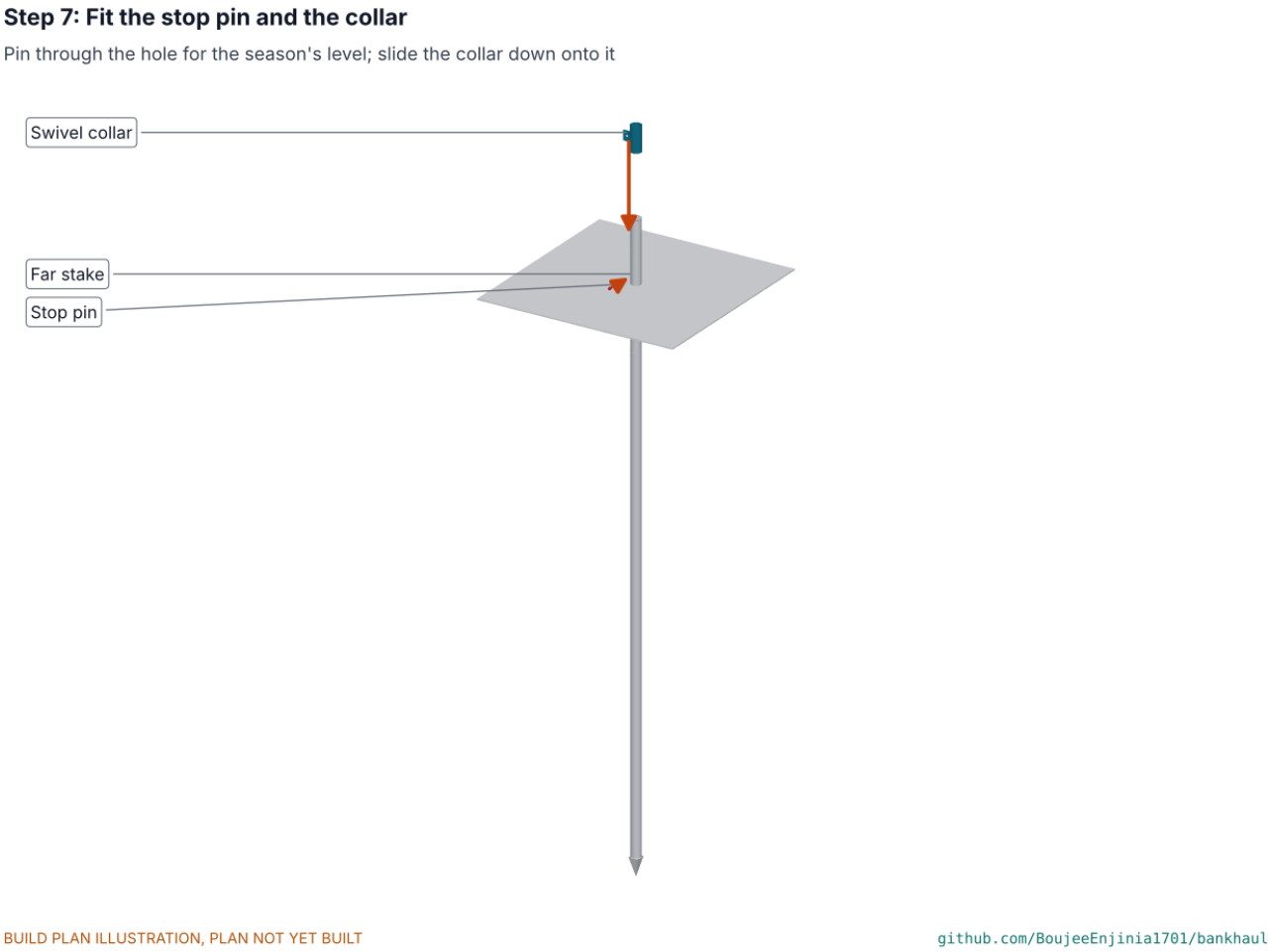
Shackle through the pad eye and the block's swivel eye; wire the pin.

Step 6: Drive the far stake from a boat



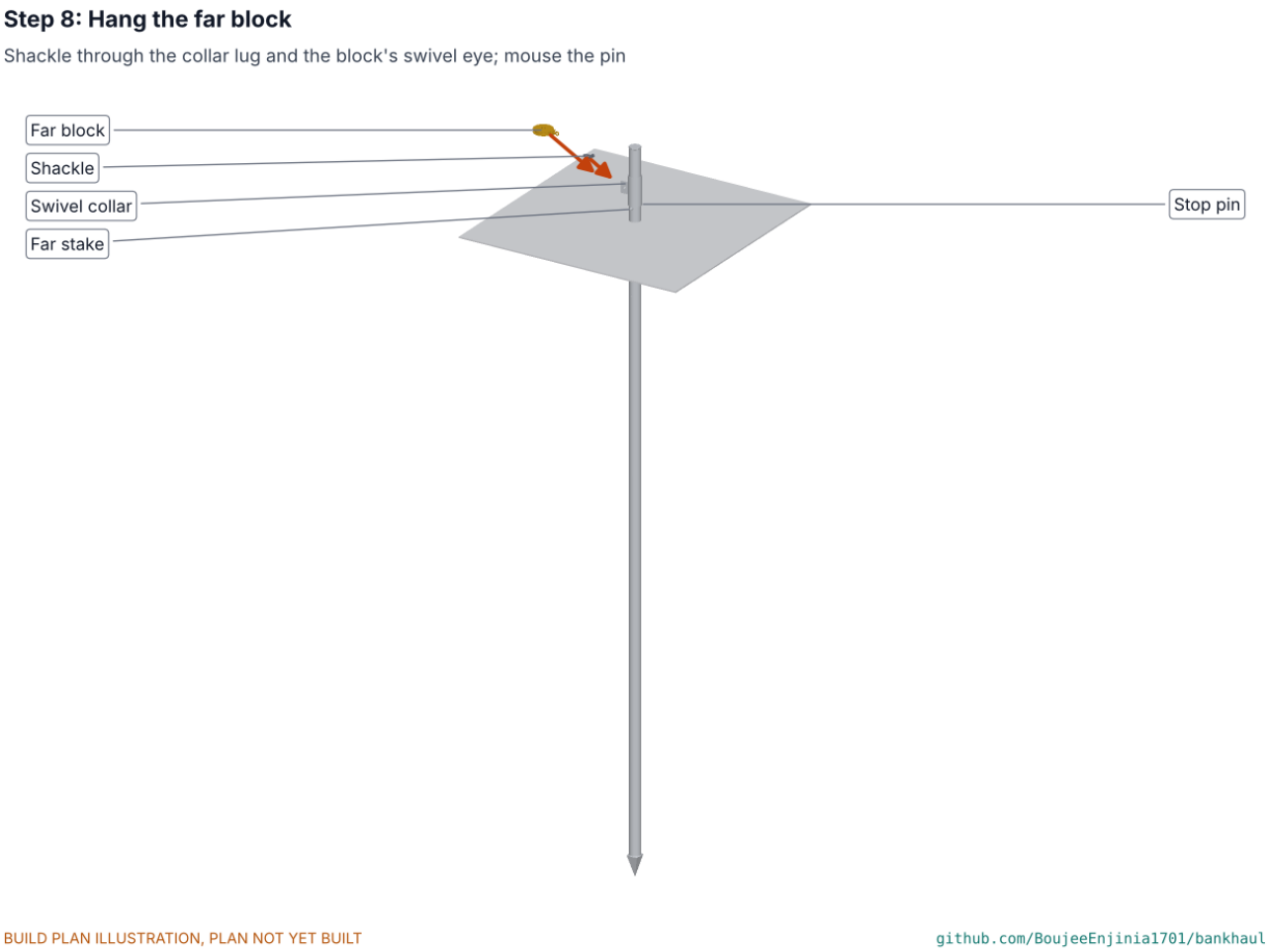
From a boat held on two anchors, or standing on the dry bed at low water, stand the stake on its point 30 m from the post on the line between the anchors. The stake is 5.0 m long and weighs 33.2 kg, so nobody holds it upright at arm's length over the side: pass the 10 m offcut of the loop coil (line 6) through the stake's top cross hole, rest the stake against the boat's side with its point down, and let it down onto its point on the rope, two people on the rope, one of them steadying the stake below the top (BKH-DEC-001 O-7). Pull the rope out of the hole once the stake stands. In soft mud it sinks part of the way under its own weight; push and twist it down as far as it will go. Then fit the driving cap and drive with a post driver or sledge until the paint ring, 3.5 m above the point, reaches the bed. Keep it upright within 2 degrees.

Step 7: Fit the stop pin and the collar



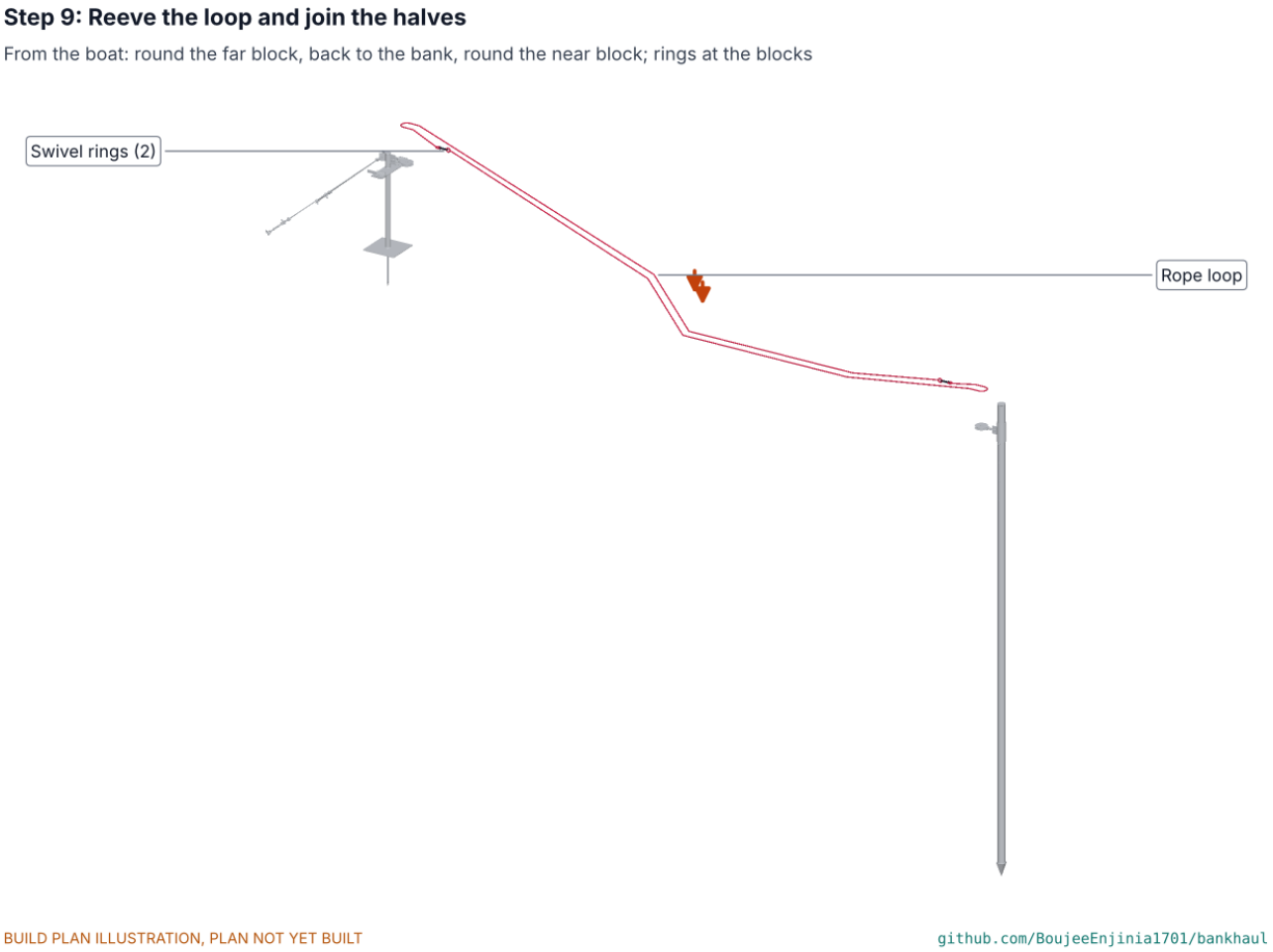
Choose the hole that puts the far block about 200 mm above the day's water. Where the bed's vane reading is under 10 kPa, never use the top hole: if the water at the stake is deeper than 1.0 m, take the loop out (D-S7). Fit the bolt and nyloc nut through it, then slide the collar down the stake onto the pin.

Step 8: Hang the far block



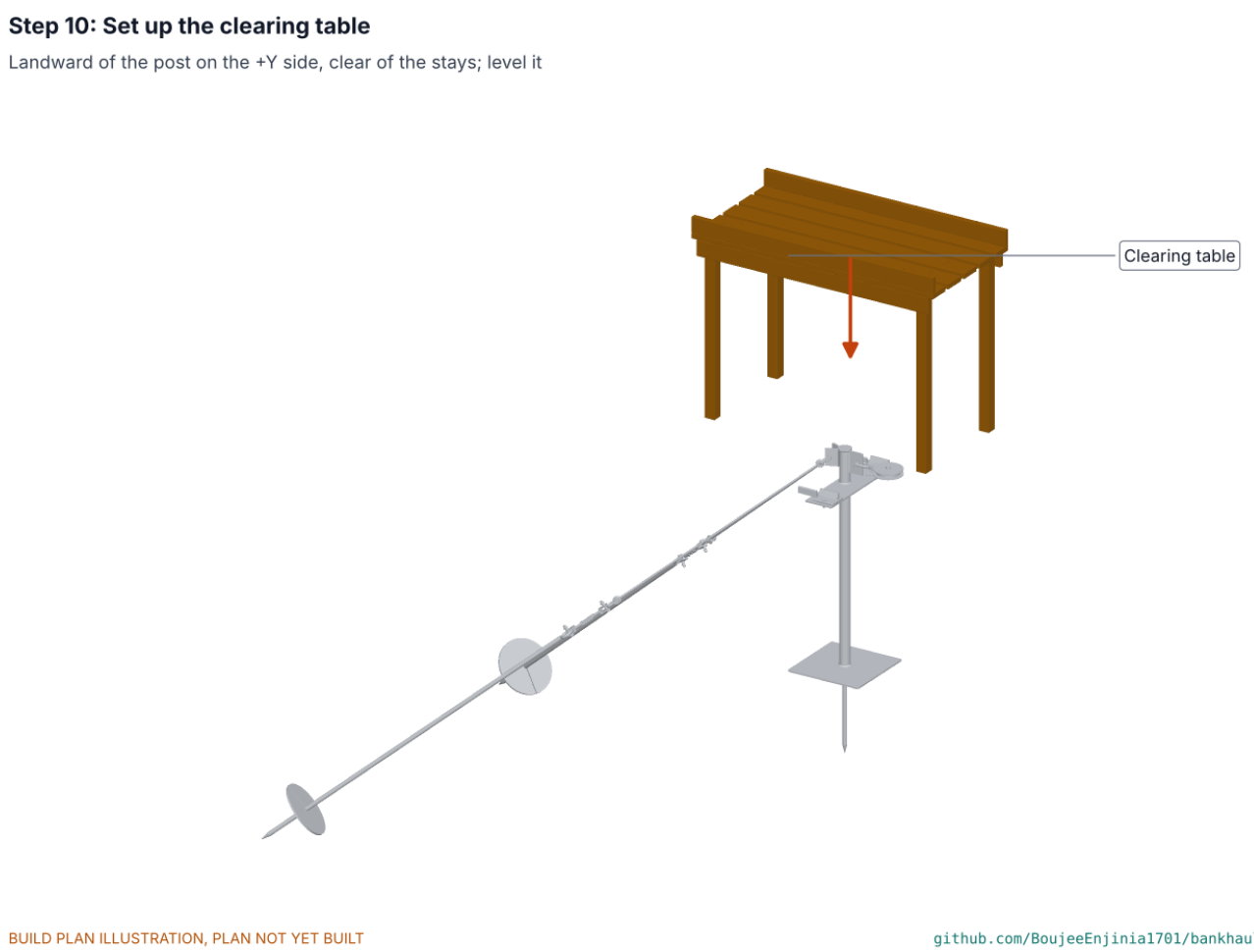
Shackle through the collar lug and the block's swivel eye; wire the pin.

Step 9: Reeve the loop and join the halves



From the boat, run the first half from the bank round the far block and back to the bank; run the second half round the near block. Splice each plain end round the other half's ring, five tucks, so that ring 1 lies about 450 mm in front of the near block when ring 2 lies about 450 mm short of the far block and the loop is snug but not tight. Coil and seize any surplus at the ring.

Step 10: Set up the clearing table

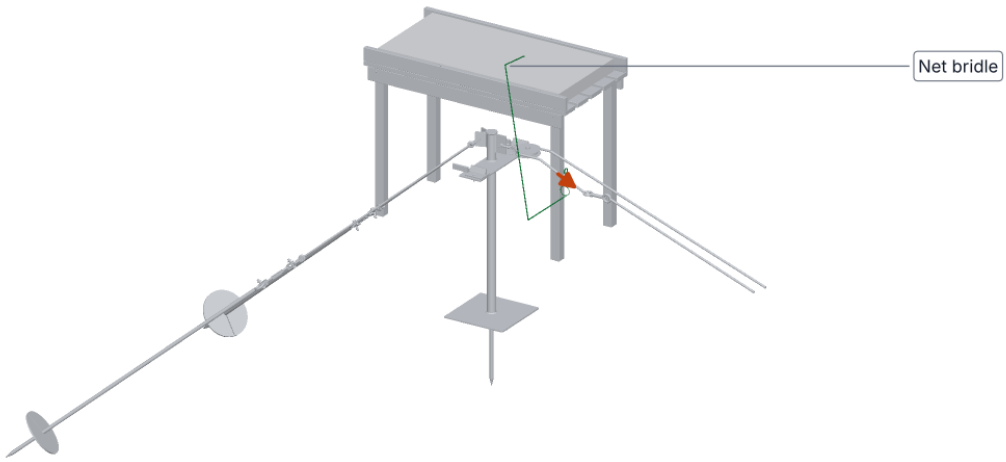


Landward of the post on the side away from the stays, level on firm ground. Then cut a 25 mm deep recess in the bank lip where the two loop legs cross the edge and set the bank-edge board in it, flush with the ground and its rounded nose at the water's edge (line 18).

Step 11: Clip the net to ring 1

Step 11: Clip the net to ring 1

Far bridle to ring 1 through its weak link; near bridle to the tie-off hole



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/bankhaul

Stack the net on the table, far end on top. Clip the far-end bridle to ring 1 and the near-end bridle to the tie-off hole. The station is now ready for the first checks.

5. First checks

Check	Requirement	How	Pass when
Post and stays	R4	Pull the near block toward the water with a spring scale or come-along to 3 kN, from behind a screen	No visible movement of post, anchors or stays; no change at the anchor eyes
Far stake	R5	Pull the collar toward the bank to 1.5 kN from the boat	The stake moves less than 25 mm at the collar and springs back
Weak link	R10	Pull three links from the batch	Each releases between 500 and 700 N
Loop runs	R2	Pull each leg by hand over a full cycle	The loop runs freely; no ring reaches a block; nothing rubs a cheek
Set and haul	R1, R3	Set and haul a 30 m net from the bank with a spring scale on the hauling end, once with the bank-edge board in place and once with it lifted out (O-2)	Nobody enters the water; peak pull under 250 N; the two peak pulls recorded
Timing	R6	Time a haul and reset	Under 10 minutes without clearing fish
Working spot	R11	Measure from where the fisher stands to the water's edge	5 m or more

6. Safety stops

- **Before driving the far stake:** two people, a boat held on two anchors or a dry bed, a lookout on the water, and the local wildlife authority's advice on the site. Never wade. The 5.0 m stake is lifted upright by two people and never driven from a standing position higher than the boat's seat.
- **Before the first load on the bank station:** stays tensioned, all shackle pins wired, anchors in line with the stays, and nobody in line with a stay or in the bight of the loop.
- **Before the first pull test:** the person pulling stands behind a screen or to the side of the line; the load is applied slowly and held, never jerked.
- **Before the first net is set:** both weak links tested from the same batch and marked; the jam cleats hold a leg against a firm pull.
- **Before each use:** the loop, splices, stays, shackles and blocks are looked over; any cut, melted or worn rope is replaced.
- **When floods are expected:** the loop is taken out of the water and the net is not set.

7. Tools, skills and workspace

- **Welding and drilling:** MIG or stick welder for 3 to 8 mm steel, angle grinder, pillar drill or magnetic drill to 49 mm, a flat table to square the post. A competent welder.
- **Woodwork:** saw, drill and screwdriver.
- **Rope work:** splicing fid, sharp knife, tape and lighter; someone who can make an eye splice in three-strand rope.

- **Site:** two people, a tommy bar for the screw anchors, a post driver or 5 kg sledge with the driving cap, a tape measure, a boat that two people can work from, life jackets.
- **Testing:** a spring scale or load cell to 5 kN and a come-along, or the CalRig proof-load rig.

8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py`
- General arrangements: `cad/drawings/BKH-DWG-001` (bank station), `BKH-DWG-002` (far end) and `BKH-DWG-003` (layout)
- Making sketches: `cad/drawings/BKH-DWG-101` to `BKH-DWG-107`; pictures: `cad/src/build_plan_media.py`
- Calculations: `docs/04-calcs/01-sizing.md` and `docs/04-calcs/sizing.py` (BKH-CAL-001)
- Parts: `bom/bom.csv`
- Design changes: `docs/decisions/0002-design-for-construction.md` (BKH-DDR-002) and `docs/decisions/0003-amish-requirement-decisions.md` (BKH-DDR-003)